

SIMPLE PROBLEM

Problem Statement:

The World Economic Forum wants a quick summary of countries where the **GDP value exceeds 1 trillion**.

MySQL Workbench Solution:

```
SELECT country_id, year_id, gdp_value  
FROM GDP  
WHERE gdp_value > 1000000000000;
```

SQLite Online Solution:

```
SELECT country_id, year_id, gdp_value  
FROM GDP  
WHERE gdp_value > 1000000000000;
```

MEDIUM PROBLEMS

Problem Statement 1:

The UN population division wants to identify which countries had the **highest GDP per capita** in 2021 and list the **top 5 countries** by this metric.

MySQL Workbench Solution:

```
SELECT
    c.country_name,
    g.gdp_value,
    p.population_value,
    ROUND(g.gdp_value / p.population_value, 2) AS GDP_per_capita
FROM GDP g
JOIN Population p
    ON g.country_id = p.country_id AND g.year_id = p.year_id
JOIN Country c
    ON c.country_id = g.country_id
WHERE g.year_id IN (
    SELECT year_id FROM Year WHERE year = 2021
)
ORDER BY GDP_per_capita DESC
LIMIT 5;
```

SQLite Online Solution:

```
SELECT
    c.country_name,
    g.gdp_value,
    p.population_value,
    ROUND(g.gdp_value * 1.0 / p.population_value, 2) AS
GDP_per_capita
FROM GDP g
JOIN Population p ON g.country_id = p.country_id AND g.year_id =
p.year_id
JOIN Country c ON c.country_id = g.country_id
WHERE g.year_id = (
    SELECT year_id FROM Year WHERE year = 2021
)
ORDER BY GDP_per_capita DESC
LIMIT 5;
```

Problem Statement 2:

A global energy study wants to find all countries whose **electricity access** was **above the world average** for the same year.

MySQL Workbench Solution:

```
SELECT
    c.country_name,
    e.electricity_percentage,
    (SELECT AVG(e2.electricity_percentage)
     FROM ElectricityPercent e2
     WHERE e2.year_id = e.year_id) AS world_avg
FROM ElectricityPercent e
JOIN Country c ON c.country_id = e.country_id
JOIN Year y ON y.year_id = e.year_id
WHERE e.electricity_percentage > (
    SELECT AVG(e3.electricity_percentage)
    FROM ElectricityPercent e3
    WHERE e3.year_id = e.year_id
)
ORDER BY e.electricity_percentage DESC;
```

SQLite Online Solution:

```
SELECT
    c.country_name,
    e.electricity_percentage,
    (SELECT AVG(e2.electricity_percentage)
     FROM ElectricityPercent e2
     WHERE e2.year_id = e.year_id) AS world_avg
FROM ElectricityPercent e
JOIN Country c ON c.country_id = e.country_id
JOIN Year y ON y.year_id = e.year_id
WHERE e.electricity_percentage > (
    SELECT AVG(e3.electricity_percentage)
    FROM ElectricityPercent e3
    WHERE e3.year_id = e.year_id
)
ORDER BY e.electricity_percentage DESC;
```

COMPLEX PROBLEM

Problem Statement:

The World Bank is conducting an advanced analysis on **rural development efficiency**. They want a report of countries for a specific year where:

1. The **GDP** is **greater than the global average GDP** for that year.
2. The **unemployment rate** is **below the global average**.
3. Each country's **rural population percentage** should be **classified** as follows:
 - **High Rural Share** → $\geq 60\%$
 - **Moderate Rural Share** → between 30% and 59%
 - **Low Rural Share** → $< 30\%$

The report must display:

country_name, year, gdp_value, unemployment_percentage, rural_population_percentage, and a computed column rural_classification.

MySQL Workbench Solution:

```
SELECT
    c.country_name,
    y.year,
    g.gdp_value,
    u.unemployment_percentage,
    r.rural_population_percentage,
    CASE
        WHEN r.rural_population_percentage >= 60 THEN 'High Rural
Share'
        WHEN r.rural_population_percentage BETWEEN 30 AND 59 THEN
'Moderate Rural Share'
        ELSE 'Low Rural Share'
    END AS rural_classification
FROM GDP g
JOIN UnemploymentPercent u
    ON g.country_id = u.country_id AND g.year_id = u.year_id
JOIN RuralPopulation r
    ON g.country_id = r.country_id AND g.year_id = r.year_id
JOIN Country c
    ON g.country_id = c.country_id
JOIN Year y
    ON g.year_id = y.year_id
WHERE g.gdp_value > (
```

```

        SELECT AVG(g2.gdp_value)
        FROM GDP g2
        WHERE g2.year_id = g.year_id
    )
    AND u.unemployment_percentage < (
        SELECT AVG(u2.unemployment_percentage)
        FROM UnemploymentPercent u2
        WHERE u2.year_id = u.year_id
    )
ORDER BY y.year, g.gdp_value DESC;

```

SQLite Online Solution:

```

SELECT
    c.country_name,
    y.year,
    g.gdp_value,
    u.unemployment_percentage,
    r.rural_population_percentage,
    CASE
        WHEN r.rural_population_percentage >= 60 THEN 'High Rural
Share'
        WHEN r.rural_population_percentage BETWEEN 30 AND 59 THEN
'Moderate Rural Share'
        ELSE 'Low Rural Share'
    END AS rural_classification
FROM GDP g
JOIN UnemploymentPercent u
    ON g.country_id = u.country_id AND g.year_id = u.year_id
JOIN RuralPopulation r
    ON g.country_id = r.country_id AND g.year_id = r.year_id
JOIN Country c
    ON g.country_id = c.country_id
JOIN Year y
    ON g.year_id = y.year_id
WHERE g.gdp_value > (
    SELECT AVG(g2.gdp_value)
    FROM GDP g2
    WHERE g2.year_id = g.year_id
)
AND u.unemployment_percentage < (
    SELECT AVG(u2.unemployment_percentage)
    FROM UnemploymentPercent u2
    WHERE u2.year_id = u.year_id
)
ORDER BY y.year, g.gdp_value DESC;

```